FIT3155: Week 11 Lab Sheet

(Questions are based on Week 10 topic dealing with Linear Programming.)

Objectives: Develop a basic understanding of solving Linear Programs in the standard form using Danzig's simplex algorithm.

Conceptual exercises

1. Consider a standard-form linear program with N nonnegative decision variables and m linear ($\leq$) constraints.

   - After introducing slack variables. How many variables are there in total?
   - How many variables are basic (free) and how many are nonbasic (fixed to 0)?
   - Characterize the size of the search space an exhaustive enumeration approach of solving a linear program has to explore?

2. Lecture solves the following linear program. Revise simplex algorithm in its algebraic form, and redo the solution on paper in its tableau form.

   Maximize

   $$z = 6x + 5y$$

   subject to the constraints:

   $$\begin{aligned} x + y &\leq 5, \\ 3x + 2y &\leq 12, \end{aligned}$$

   where x and y are non-negative.

3. In each iteration of the simplex method, as you would have revised above, a non-basic variable enters the basis (becoming a basic variable in the next iteration), replacing a basic variable that leaves the basis (becoming non-basic in the next iteration).

   (i) How is the choice of the entering non-basic variable made in simplex?

(ii) How is the choice of the leaving basic variable made in simplex?

(iii) What is the stopping criterion of the simplex algorithm?

4. On paper, solve the following linear program using the tableau simplex method:

   Maximize

   $$z = x + 2y$$

   subject to the constraints:

   $$\begin{aligned} 4x + y &\le 44, \\ 3x + 2y &\le 39, \\ 2x + 3y &\le 37, \\ y &\le 9, \\ -x + y &\le 6 \end{aligned}$$

   where x and y are non-negative.

   Once solved, report the values of the decision variables that maximize the objective function.

5. On paper, solve the following linear program using the tableau form of the simplex method:

   Maximize

   $$z = x + 2y$$

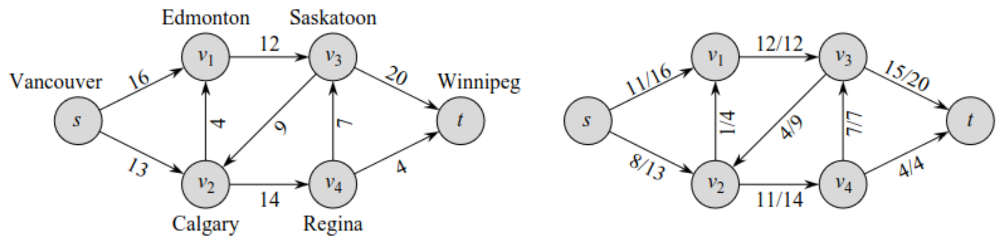
   subject to the constraints:

   $$\begin{aligned} 2x + 4y &\le 12, \\ 4x + 3y &\le 16, \end{aligned}$$

   where x and y are non-negative.

6. In the above linear program what was the stopping criteria? Is the solution you stopped at unique? Justify your answer.

7. In FIT2004 you have covered the maximum flow problem. Recall that we are given a directed graph $G = (V, E)$ in which each edge $(u, v) \in E$, with $u, v \in V$, has a nonnegative capacity $c(u, v) \ge 0$, and two special vertices $s, t \in V$ designated as the source and sink, respectively. (Assume that there are no incoming edges into the source s and no outgoing edges from the sink t.)



(Figure source: CLRS Fig 26.1)

A flow in the network assigns a value $f(u, v)$ to each edge $(u, v) \in E$ and satisfies the capacity constraint on each edge in E, and flow conservation constraint at every vertex $v \in V \setminus \{s, t\}$. The value of a flow is defined as the total flow leaving the source s. A maximum flow is a feasible flow that maximizes the flow value.

Based on these variables, express the maximum flow problem as a linear program.

(Note: This formulation will not be in standard form because some constraints are equalities. You don't have to solve it as solving linear programs in non-standard forms is non-examinable. This question, nevertheless, will be useful to appreciate that the network flow problem is a special case of linear programming. In fact, the Ford-Fulkerson algorithm (and its variants/specializations) starts from the flow value of 0 and repeatedly increases the flow until no further improvement is possible. When the flow cannot be increased any further and terminates, the flow value is maximized, which follows the behaviour of the simplex method. In fact, even shortest-distance problem for which you have learnt specialized algorithms can be converted into a linear program and solved.)

8. Write pseudocode for the tableau simplex method to solve a linear program in standard form. Assume inputs:

   (i) A vector c containing the coefficients of the objective function.
   - For example, for the linear program in Q2: $c = [6, 5, 0, 0]$.

   (ii) A matrix B containing the coefficients of the LHS of the basic equations.
   - For example, for the linear program in Q2:

$$B = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{bmatrix}$$

   (iii) A vector rhs containing the RHS of the basic equations.
   - For example, for the linear program in Q2: $rhs = [5, 12]$.

-=0=-
END
-=0=-